Sora Kim

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Research Focus

Responsibility is produced, not assigned. My research investigates **accountability infrastructure in AI-integrated regulated work**—how regulated organizations structure responsibility in the first place, and how AI integration reshapes that production. I combine organizational theory, science and technology studies (STS), design research, and computational text analysis, grounded in nearly four years of embedded practitioner-research in pharmaceutical GMP quality control, where I have deployed 9 production tools serving a ~50-person quality team.

Keywords: accountability infrastructure · AI adoption in organizations · technology & innovation management · regulated work · judgment preservation · sociotechnical systems

Education

Chungbuk National University *Cheongju, South Korea*
B.A. in English Literature and Linguistics March 2015 – August 2022

- **Top-Ranked Admission**, Humanities & Social Sciences Division, 2015 (entrance examination).
- **Full Merit Scholarship** (tuition & fees, 8 semesters; merit-based).

Self-Directed Study *Remote*
Programming, Data Science, Machine Learning March 2021 – August 2022

- Completed 7-course specialization in data science and computational methods (Johns Hopkins University, Coursera, 2021), including coursework in algorithms, statistics, Python, R, and command-line tools applied to large-scale datasets—foundation for later text-corpus analysis and computational work in the research program.
- Applied deep learning projects: image classification (PyTorch, CNN); unsupervised learning; ensemble methods.

Industry Research Experience

GMP-Regulated Pharmaceutical Manufacturing Site *Seoul, South Korea*
Quality Control / Quality Systems September 2022 – Present

- Embedded practitioner-researcher in regulated biomanufacturing quality operations (nearly four years as of application cycle).
- Conducted sustained ethnographic observation of Standard Operating Procedure (SOP) use, regulatory deviation handling, Corrective and Preventive Action (CAPA) processes, and regulatory audit workflows.
- Designed and deployed 9 productivity tools serving Quality Control (QC) operations (see Deployed Systems).
- Designed and delivered a differentiated AI training program for the QC team (managers: tool-building; operators: AI-assisted authoring), establishing the team's first structured reference for AI in GMP workflows; observations from this program inform ongoing research on AI integration in regulated environments.
- Observational data and fieldwork form the empirical foundation of the research program below.

Deployed Systems *(Empirical Substrate for Design Research)*

Production tools designed and deployed at a GMP-regulated pharmaceutical manufacturing site. All systems operate as unvalidated productivity layers external to validated GMP systems, with team lead awareness and support. These systems are not peripheral artifacts but the empirical substrate of the research program—each tool serves as a site of ethnographic observation, behavioral data collection, and iterative design, and as a concrete test case for how AI can be integrated into regulated work without eroding accountability structures.

- **QC Stability Testing Dashboard** — Deadline-tracking interface for stability testing programs; team-wide adoption, ongoing operation.
- **SOP Retrieval System** — Semantic search tool for GMP procedural documents; empirical base for *From Retrieval to Reasoning* (see Publications).
- **Document Navigation Tool (Traceability Layer)** — Cross-document linking interface with metadata mapping across regulated quality documentation; empirical base for *When Documents Don't Connect* (traceability component).
- **Navigation Behavior Tracker** — Behavioral tracking layer capturing navigation paths, hesitation, and cross-reference jumps in regulated document work; empirical base for *When Documents Don't Connect* (loopback component).
- **Review Dashboard** — Prioritized document review interface surfacing procedural friction before formal deviation; empirical base for *Friction Before Failure*.
- **Root Cause Analysis (RCA) Scaffolder** — Investigative scaffold that preserves analyst judgment while structuring causal reasoning; empirical base for *Keeping Comparison Open*.
- **Veeva Vault Change Control Assistant** — Workflow support for electronic change control submissions.
- **Reagent Inventory & Expiry Management Tool** — Automated expiry tracking and inventory alerts for QC reagents.
- **Document Auto-Review AI** — Format and compliance pre-check for procedural docu-

ments.

Publications

Organizational Theory & STS — Under Review

A four-paper lifecycle theory of accountability in regulated organizations: pre-action structuring (AAC), procedural document infrastructure (CF), investigative convergence (RC), and AI-accelerated closure (PAC). The lifecycle articulates how responsibility is produced structurally at each stage, with implications for AI integration in regulated work.

Kim, S. (2026). *Anticipated Accountability Convergence: How Organizations Structure Responsibility Before Action.* Under review at **Academy of Management Review** (AMR-2026-0516). SSRN: 6371980

Kim, S. (2026). *Cognitive Friction: Procedural Documents as Accountability Infrastructure in Regulated Organizations.* Under review at **Organization** (ORG-26-0241). SSRN: 6391278

Kim, S. (2026). *Responsibility Convergence: How Investigative Systems Stabilize Attribution.* Under review at **Organization Science** (ORSC-MS-2026-22511). SSRN: 6372301

Kim, S. (2026). *Premature Accountability Convergence: How Technology Accelerates Responsibility Stabilization in Organizations.* Under review at **Information and Organization** (INFORG-D-26-00222). SSRN: 6391379

Design Research — In Preparation

Five design studies articulating a four-layer framework—accessibility, structure, coordination, reasoning—for accountability infrastructure in AI-integrated regulated work. Each study is grounded in a deployed production tool (see Deployed Systems) and anchored in the three design principles of upstream stabilization, observability over optimization, and judgment preservation.

Kim, S. (2026). *Accountability Infrastructure Design: A Four-Layer Framework for Judgment-Preserving Systems in Regulated Work.* In preparation for submission to **ACM Transactions on Computer-Human Interaction (TOCHI)**.

Kim, S. (2026). *Keeping Comparison Open: Interface Scaffolding Against Premature Accountability Convergence in Regulated Investigations.* In preparation for submission to **CSCW 2027**.

Kim, S. (2026). *Friction Before Failure: Making Review Burden Visible in Regulated Document Work.* Working paper (SSRN).

Kim, S. (2026). *From Retrieval to Reasoning: How SOP Accessibility Shapes Interpretation in Regulated Work.* Working paper (SSRN).

Kim, S. (2026). *When Documents Don't Connect: Traceability Burden and Loopback Behavior in GMP Document Navigation.* Working paper (SSRN).

Computational Research — Working Papers

A computational research program operationalizing accountability across four signal layers (detection, graph structure, attribution, distribution), applied to FDA Warning Letters and synthesized in the umbrella paper. These analyses provide external validation of the lifecycle theory through

large-scale regulatory text.

Kim, S. (2026). *Responsibility as Multi-Layer Signal: A Computational Framework for FDA Warning Letters.* Working paper (SSRN). [Umbrella]
Kim, S. (2026). *Premature Convergence Detection in Regulatory Investigative Texts.* Working paper + GitHub: sorakim-lab/premature-convergence-detection.
Kim, S. (2026). *Accountability Graph Extraction from Regulatory Texts.* Working paper (SSRN).
Kim, S. (2026). *Responsibility Attribution Prediction: A Classification Approach to FDA Warning Letters.* Working paper (SSRN).
Kim, S. (2026). *Responsibility Distribution Metric: An Entropy-Centrality Approach to Accountability Structure.* Working paper (SSRN).

Conferences

- Kim, S. (2026). *Premature Accountability Convergence: Inscription, Closure Pressure, and the Stabilization of Responsibility.* Abstract submitted to **4S (Society for Social Studies of Science) Annual Meeting**, Toronto, 2026 (Panel 74).

Research Corpus & Data Infrastructure

- **FDA Warning Letters Corpus** — N=50 letters, 160 violations (2016–2025). Used across organizational theory, STS, and computational papers.
- **Cognitive Friction Corpus** — 400 naturally occurring sentences from regulated procedural documents, manually annotated at sentence level.
- **Field Observation Archive** — Nearly four years of notes on SOP use, deviation handling, and CAPA in a GMP-regulated site.

Methods & Technical Skills

Research Methods: Ethnographic fieldwork; document & corpus analysis; within-subject experimental design; think-aloud protocols; behavioral logging; qualitative coding; annotation protocol design; causal reasoning analysis; statistical analysis.

Programming & Development: Python, R; Streamlit (deployment); Git/GitHub; LaTeX; HTML/CSS.

Text Analysis & NLP: TF-IDF; sentence-level annotation; document classification; corpus construction; qualitative coding.

Machine Learning: PyTorch; scikit-learn; classification; ensemble methods; unsupervised learning.

Regulated-Environment Systems: Veeva Vault QMS; GMP documentation systems; electronic change control; deviation/CAPA workflows.

Languages

Korean (native) · English (fluent) · Japanese (fluent)